NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 12-04-26**
Time: 3 Hrs **Max. Marks: 180**

12-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-5(P1)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 5 – 10)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 11 – 16)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 23)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 24 – 29)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 30 – 35)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 42)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 43 – 48)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 49 – 54)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				**19**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Let O be the centre of the circle $x^2+y^2=r^2$, where $r>\sqrt{\frac{5}{2}}$. Suppose, PQ is a chord of this circle and the equation of the line passing through P and Q is $2x+4y=5$. If the centre of the circumcircle of ΔOPQ lies on the line $x+2y=4$, then the value of r is

A) 3 B) 2 C) 5 D) 9

2. Suppose a is a real number such that the equation

$a.(\sin x+\sin(2x))=\sin(3x)$

has more than one solution in the interval $(0,\pi)$. Number of integral values of 'a' satisfying the given condition is

A) 0 B) 1 C) 2 D) 3

3. Two tangents, one from the point $A(2,1)$ and other from the point $B(-2,1)$ are drawn to the hyperbola $\frac{x^2}{4}-\frac{y^2}{1}=1$. A circle which touches these two tangents and two asymptotes to the hyperbola has its centre at (a,λ) where $\lambda>0$

The least area of the pentagon in which the above circle is inscribed (two sides of the pentagon are asymptotes of the hyperbola and two sides are the here tangents), is

A) $\left(3+\sqrt{5}\right)sq.units$ B) $\left(3+2\sqrt{5}\right)sq.units$

C) $\left(6+2\sqrt{5}\right)sq.units$ D) $\left(6+4\sqrt{5}\right)sq.units$

4. The total number of positive integral solution of $15<x_1+x_2+x_3\le 20$ is equal to

A) 685 B) 785 C) 1125 D) 1200

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SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 5 and 6:

Given set of vectors:

$$\vec{a}=\frac{-2}{3}t^3\hat{i}-\frac{3}{2}t^2\hat{j}+\left(\frac{\lambda}{3}-10\right)\hat{k},$$

$$\vec{b}=\hat{i}+2\hat{j}+3\hat{k},$$

$$\vec{c}=4\hat{i}+5\hat{j}+6\hat{k},$$

t is a real number which satisfies such that given thee vectors are linearly dependent vectors, and $t=x+\frac{1}{x}$ and $\lambda\in R$ then

5. The value of λ, for which the given equation will have three real and distinct solutions of x is ____

6. If the given equation has exactly two real and distinct solution(s) of x, then the number of integral values of λ in $[-25,15]$ is _____

Question Stem for Question Nos. 7 and 8:

ABCD is a tetrahedron having each edge of unit length

7. The volume of the tetrahedron is $\frac{1}{K\sqrt{2}}$. Then K is

8. If the angle between two adjacent faces is $\cos^{-1}\left(\frac{1}{K}\right)$, then $K=$ _____

Question Stem for Question Nos. 9 and 10:

Consider the system of equations $\sin x\cos 2y=\left(a^2-1\right)^2+1,\ \cos x\sin 2y=a+1$

9. The number of values of a for which the system has a solution is

10. The number of values of $x\in[0,2\pi]$, when the system has solution for permissible values of a, is/are

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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

11. Let $k=\int_0^2 x^2\left(x^2+2\right)^{\frac{1}{5}}dx$ then, which of the following is incorrect

A) $[k]=2$ (where [ ] denotes Greatest integer function)

B) $3<k<4$

C) k is a prime number

D) k is an odd number

12. If $S={}^{10}C_0\ {}^{20}C_{10}-{}^{10}C_1\ {}^{18}C_{10}+{}^{10}C_2\ {}^{16}C_{10}-{}^{10}C_3\ {}^{14}C_{10}+{}^{10}C_4\ {}^{12}C_{10}-{}^{10}C_5\ {}^{10}C_{10}$, then find the incorrect statement

A) Number of divisors of S is 10

B) Sum of digits of S is 6

C) S is a four digit number

D) Number of divisors of S is 9

13. Let $f:R\to R$ be a function such that $f\left(x^5+x-2\right)=x^3-1,\ \forall x\in R$ and g be the inverse function of f.

If $p=\frac{d}{dx}\left(\left(x^3+1\right)\times g\left(f^3\left(g\left(f\left(g(x)\right)\right)\right)\right)\right)_{at\ x=7^{1/3}}$, $q=\frac{d^2}{dx^2}\left(\frac{f\left(f\left(g\left(g\left(f(x)\right)\right)\right)\right)}{x^2}\right)_{at\ x=-2}$

and $N=\left(p\left(7^{-\frac{2}{3}}\right)+8q\right)$, then (here $f^n(x)=\left(f(x)\right)^n$)

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A) Sum of the digits of N is 12

B) N is divisible by 5

C) Number of positive divisors of N is 4

D) N contains three distinct digits

14. Let z_1, z_2, z_3 be the three nonzero complex numbers such that $z_2 \neq 1$, $a = |z_1|$, $b = |z_2|$ and $c = |z_3|$. Let $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$. Then

A) $\arg\left(\frac{z_3}{z_2}\right) = \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)^2$

B) Orthocenter of triangle formed by z_1, z_2, z_3 is $z_1 + z_2 + z_3$

C) If triangle formed by z_1, z_2, z_3 is equilateral, then its area is $\frac{3\sqrt{3}}{2}|z_1|^2$

D) If triangle formed by z_1, z_2, z_3 is equilateral, then $z_1 + z_2 + z_3 = 0$

15. Two functions $f(x)$ and $g(x)$ are defined as $f(x) = \begin{cases} [x]^{\{x\}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $g(x) = \begin{cases} \{x\}^{[x]}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, (where [.] and {.} refer to the greatest integer function and the fractional part function respectively). Then which of the following statements is/are true (where $n \in N$)

A) $\int_0^{n+1} f(x)\,dx - \int_0^n f(x)\,dx = \frac{n-1}{\ln n}$

B) $\int_0^{n+1} g(x)\,dx - \int_0^n g(x)\,dx = \frac{1}{n+1}$

C) $\int_0^t f(x)\,dx = \int_0^t g(x)\,dx$ has at least one solution in $t \in \left(0, \frac{\pi}{2}\right)$

D) $e < \lim_{n\to\infty} \int_0^n g(x)\,dx < e^2$

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16. Let α be the region in the plane $x+y+z=1$ with $x,y,z\geq 0$. β is set of points (a,b,c) in α such that only two of the three inequalities: $a\leq\frac{1}{2},b\leq\frac{1}{3},c\leq\frac{1}{6}$ hold simultaneously. If area of region R is denoted by $A(R)$, then

A) $A(\alpha)=\frac{\sqrt{3}}{4}$ B) $A(\alpha)=\frac{\sqrt{3}}{2}$ C) $A(\beta)=\frac{7\sqrt{3}}{36}$ D) $A(\beta)=\frac{11\sqrt{3}}{36}$

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

17. The equation of transverse axis of hyperbola (passing through origin) having asymptotes $3x-4y-1=0$ and $4x-3y-6=0$ is $ax+by-c=0,\ a,b,c\in N$ and $\gcd(a,b,c)=1$, then the value of $(a+b+c)=$ _____

18. If $P(x)$ be a polynomial satisfying $P(x)-2P'(x)=3x^3-27x^2+38x+1$. If function $f(x)=\begin{cases}\frac{P''(x)+18}{6}, & x\neq\frac{\pi}{2}\\ \sin^{-1}(ab)+\cos^{-1}(a+b-3ab), & x=\frac{\pi}{2}\end{cases}$ is continuous at $x=\frac{\pi}{2}$, then $(a+b)$ is equal to _____

19. The units digit in the integral part of the expansion of $\left(15+\sqrt{220}\right)^{19}\left(1+\left(15+\sqrt{220}\right)^{63}\right)$ is _____

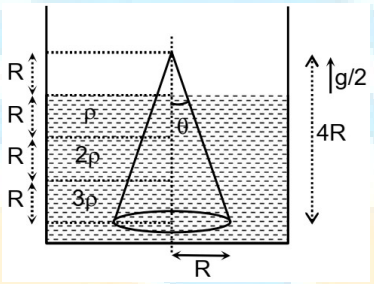
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PHYSICS **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

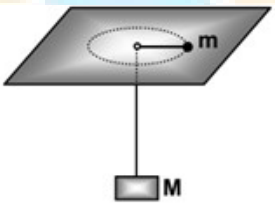
This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

20. A solid cone is kept in three immiscible liquids of density ρ, 2ρ and 3ρ inside a container. The radius of cone is 'R' and its height is 4R. Container is accelerated with a constant acceleration g/2 in vertically upward direction. If cone is in equilibrium relative to the frame of container, then density of the cone is



A) $\frac{73}{24}\rho$ B) $\frac{146}{25}\rho$ C) $\frac{39}{16}\rho$ D) $\frac{146}{37}\rho$

21. A particle of mass m is attached to one end of a light inextensible string which passes through a hole in a frictionless table top. At the other end of the string is attached a block of mass M. Initially the particle is moving in a circle of radius r with angular velocity ω. If the hanging block is pulled slightly downwards and then released, it will undergo small oscillation. Find the frequency of the oscillation.



A) $\frac{1}{2\pi}\sqrt{\frac{3m\omega^2}{M+m}}$ B) $\frac{1}{2\pi}\sqrt{\frac{3m\omega^2}{M+3m}}$ C) $\frac{1}{2\pi}\sqrt{\frac{3m\omega^2}{M}}$ D) $\frac{1}{2\pi}\sqrt{\frac{M\omega^2}{M+3m}}$

22. A pipe P open at one end but closed at the other end is filled with hydrogen gas at temperature 127^0C. Another pipe Q open at both ends and having length equal to twice the length of pipe P is filled with a mixture 1 mole of Hydrogen and 1 mole of Helium gases at temperature 147^0C. The pipe P is vibrating in its first overtone and pipe Q is vibrating in its second overtone. The ratio of their frequencies of vibration $\frac{f_P}{f_Q}$ is

A) $\sqrt{\frac{3}{5}}$ B) $\sqrt{\frac{1}{3}}$ C) $\sqrt{\frac{2}{3}}$ D) $\sqrt{\frac{4}{3}}$

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23. An asteroid orbiting around a planet in circular orbit suddenly explodes into two fragments in mass ratio 1:5. If immediately after the explosion, the smaller fragment starts orbiting the planet in reverse direction in the same orbit, what will happen with heavier fragment?

A) Fall on the planet

B) Start orbiting the planet in a larger orbit

C) Start orbiting the planet in a smaller orbit

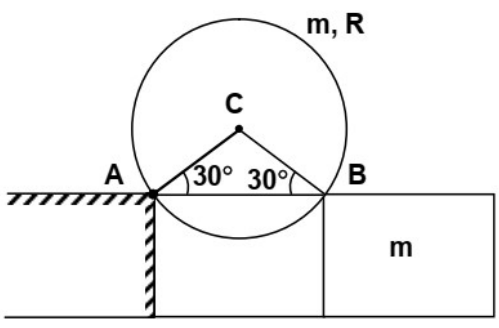
D) Escape from the gravitational interaction with the planet

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 24 and 25:

A uniform circular ring of mass $m = 5kg$ and radius $R = 30\ cm$ is hinged at the point $'A'$ of a stationary surface with the line joining point $'A'$ to the centre $'C'$ of the ring is initially vertical. The ring is also in contact with the Plank of mass $m = 5kg$ at point $'B'$ which is placed on a smooth horizontal surface. Initially points $'A'$ and B are almost touching and the line AC is vertical and the system is released from rest (there is no friction between the ring and the plank and $g = 10\ m/s^2$)



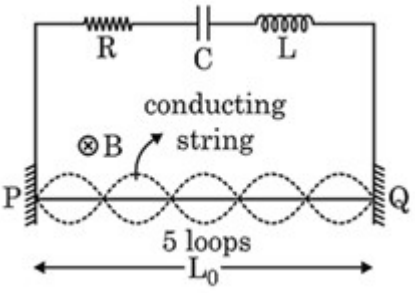
24. The speed of the plank at the instant shown is _____ m/s

25. The kinetic energy of the ring at the instant shown is _____ J

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Question Stem for Question Nos. 26 and 27:

A conducting string is rigidly clamped at the two insulated supports P and Q, separated by a distance 'L_0'. A magnetic field of uniform intensity 'B_0'$\otimes$ exists in the region. The string is given a slight disturbance due to which standing waves are set in and it vibrates in 5th harmonic. The maximum amplitude of oscillations of the string particles is "A" and the angular frequency 'ω'. It is given that at $t=0$, the particles of the string are at the mean position.

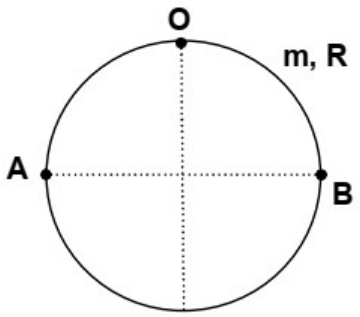


26. If the maximum value of induced Electromotive force in the string is $\frac{\alpha B_0 A\omega L_0}{\pi}$, then the value of $\alpha =$ _____

27. Assuming $\left(X_L - X_C\right)=\sqrt{3}R,$ if the magnitude of current in the circuit at $t=\frac{\pi}{2\omega}$ is $\beta\frac{B_0 A\omega L_0}{\pi R}$, then the value of $\beta =$ _____

Question Stem for Question Nos. 28 and 29:

A uniform circular ring of mass 'm' and radius 'R' is hinged at its top point 'O' and suspended in a vertical plane so that it can freely rotate about a horizontal axis passing through the hinge 'O'as shown in the figure. The bead 'A' is rigidly attached to the ring and the bead 'B' can slide without friction on the ring. The mass of each bead is 'm'. The system is released from rest from the initial position shown in the figure. Immediately after the system is released from rest, (Take $g=10m/s^2$)



28. The acceleration of bead 'A' is _____ m/s^2

29. The acceleration of bead 'B' is $2\sqrt{n}\ m/s^2$, then n =

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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

30. An imaginary particle has a charge equal to the charge of electron but mass of imaginary particle is 50 times the mass of electron. Now, consider that imaginary particle moves in circular orbit around a nucleus. The charge of nucleus is +4e, and nucleus is assumed to be at rest. Assume mass of nucleus is very large as compared to the mass of imaginary particle moving around it. Choose the correct option(s).

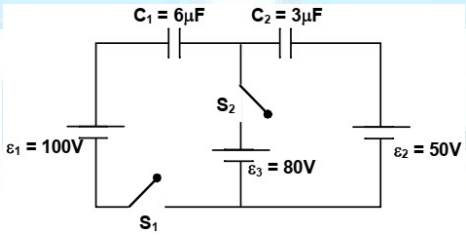
A) The radius of nth orbit is $\frac{n^2h^2\varepsilon_0}{200\pi me^2}$, where m is mass of electron, h is Planck constant, e is charge of electron.

B) If all atoms are at 16th excited state then during de-excitation till 10th excited state the total maximum number of possible wavelengths emitted is 15.

C) The wavelength of photon emitted when transition takes place from third excited state to ground state is $1.21\overset{0}{A}$.

D) Since mass of imaginary particle is different from the mass of electron so the velocity of imaginary particle and the electron in n^{th} orbit will be different.

31. In the circuit as shown in the adjoining figure the switch S_1 is closed at $t=0$. After steady state is reached, the switch S_2 is also closed for a long time. Then choose the correct option(s).



A) The heat dissipated in the circuit after the switch 'S_2' is closed will be 100 μJ.

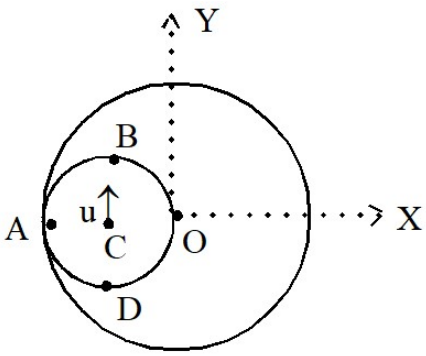
B) The heat dissipated in the circuit after the switch 'S_2' is closed will be 50 μJ

C) The charge flown through the battery ε_1 after closing the switch 'S_2' is 20 μC

D) The charge flown through the battery ε_2 after closing the switch 'S_2' is 10 μC

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32. A disc of radius R is rolling on the inner surface of a fixed coplanar ring of radius 2R. A,B,C and D are points on the disc and O is centre of ring. At $t=0$, the point A is in contact with the ring , points A and C are along x-axis and points B and D lie on a line parallel to y-axis, as shown in the figure. If speed u of the centre C of the disc is a constant, then select correct statement(s)



A) The maximum y-coordinate of point B is same as maximum y-coordinate of point DB) The maximum y-coordinate of point A is same as maximum y-coordinate of point B

C) x-coordinate of point A at any time t is given by $x=-2R\cos\left(\frac{u}{2R}t\right)$

D) x-coordinate of point A at any time t is given by $x=-2R\cos\left(\frac{u}{R}t\right)$

33. Least count (LC) is the minimum length measured by a Vernier callipers and Vernier Count (VC) = 1 MSD – 1 VSD. Main scale division (MSD) of the callipers is 1 mm and N divisions of main scale are divided into 10 Vernier scale divisions (VSD). Which of the following statement(s) is/are correct?

A) If N = 5, then LC =0.2mm and VC = 0.5 mm

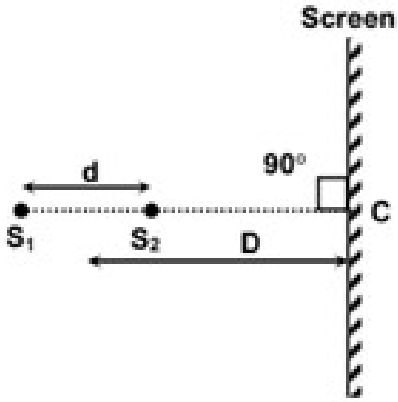
B) If N = 6, then LC = 0.2 mm and VC = 0.4 mm

C) If N = 7, then LC = 0.1 mm and VC = 0.3 mm

D) If N = 8, then LC = VC = 0.2 mm

For More Join: @IITJEE_Advanced

34. Two coherent sources are emitting light of wavelength $\lambda = 5000\overset{0}{A}$. The separation between the sources is 2.5 mm and the separation between the source and the screen is 2m. Then which of the following are correct.



A) The position of 3000th order bring fringe from C is $\frac{8}{3}m$.

B) The position of 2500th order bring fringe from C is $2\sqrt{5}\ m$.

C) The position of 5th bright fringe from C is approximately 0.08 m (including the bright fringe at C)

D) The shape of fringes on the screen are hyperbolical

35. A uniform cylinder made of material density $\rho = 125\ kg/m^3$ pressed down and held at rest in a pool so that its upper circular surface is in level with the water surface. Length of the cylinder is $\ell = 5.40\ m$ and radius is $r = 10\ cm$. When released the cylinder jumps out of water. Water is calm while releasing, the axis of the cylinder is vertical, the cylinder comes out moving vertically. Then choose the correct option(s). (density of water is $\rho_0 = 1000\ kg/m^3$ and acceleration due to gravity is $g = 10\ m/s^2$)

A) The magnitude of velocity when the cylinder just leaves the water surface is 18 m/s.

B) The maximum height above the initial position attained by the cylinder (displacement of COM of cylinder) after releasing is 16.2 m

C) The maximum height above the initial position attained by the cylinder (displacement of COM of cylinder) after releasing is 21.6 m.

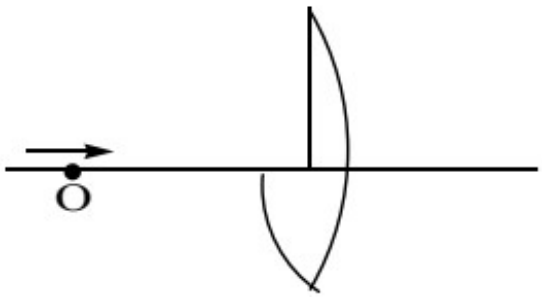
D) The loss in potential energy of liquid when the cylinder comes out from the liquid, is 1458π Joule

For More Join: @IITJEE_Advanced

SECTION - 4 (Maximum Marks : 12)

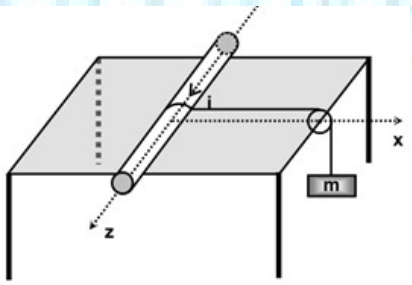
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

36. A quarter portion of a thin equi-convex lens with focal length 30 cm has been removed by cutting it perpendicular to the principal axis as shown. A small object is placed on principal axis at a large distance from lens an is moving towards the lens. At an instant, it is observed that the relative velocity between the two distinct images (corresponding to paraxial rays) is zero. The distance of object at this instant is $5\,k$ centimeters what is $'k'$?



37. Two sound waves of slightly different frequencies have amplitude ratio $\frac{11}{9}$. What is the difference of sound level in decibels of maximum and minimum intensities heard at a point:

38. A conducting rod of length $'\ell'=3$ meter and mass 4 kg lies on the horizontal table and connected with a hanging block of mass 3 kg by an ideal string passing over the pulley as shown in the figure. Coefficient of friction between the rod and the table is $'\mu'=\left(\frac{3}{4}\right)$. If the current in the conductor is 2 ampere, then find the minimum magnitude of magnetic field strength (in Tesla) such that conducting rod just starts to translate along negative $x-$axis. (take $g=10\ m/\sec^2$) [neglect the radius of rod]



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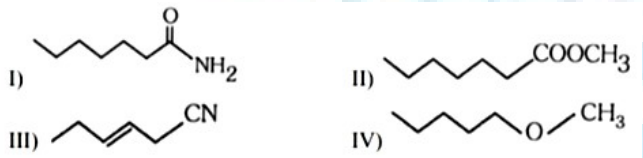
CHEMISTRY **MAX.MARKS: 60**

SECTION – 1 (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

39. The following four compounds are subjected to different reactions

Choose the correct statements for these reactions



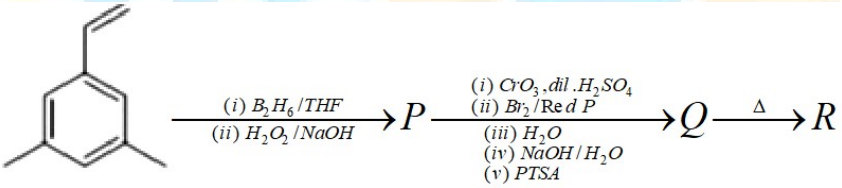
A) Compound I on reaction with bromine and $NaOH$ gives the same compound as that obtained by complete reduction of III.

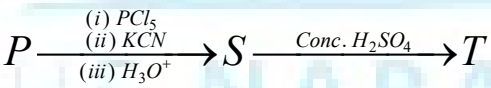
B) All compounds given I, II, III, IV undergoes reduction with $H_2 + Ni$

C) Compound III on reaction with $LiAlH_4$ gives a compound which reacts with bromine $+CCl_4$ under suitable condition gives meso compound

D) The compound obtained by the reaction of $LiAlH_4$ with II is same as that obtained on heating IV with dilute acid.

40. Consider the following reaction scheme and choose the incorrect option for the major products P, Q, R, S, and T





(PTSA: p-Toluenesulfonic acid)

A) Both P and Q turn acidified dichromate green

B) DU of R is 11

C) S gives effervescence with sodium bicarbonate

D) T gives positive iodoform test

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41. An element $X\,(At.wt.=24)$ form FCC lattice. If the edge length of lattice is 4×10^{-8} cm and the observed density is $2.4\times10^{3}\ kg/m^{3}$. Then the percentage occupancy of lattice point by element X is : $\left(N_A=6\times10^{23}\right)$

A) 96 B) 98 C) 99 D) 99.9

42. A metal complex (octahedral) having composition $Cr(NH_3)_4Cl_2Br$ have been isolated in two forms A and B. The form 'A' reacts with $AgNO_3$ to give a white precipitate which is readily soluble in dilute aqueous ammonia, while 'B' gives pale yellow precipitate which is partly soluble in concentrated ammonia. Which of the following is **incorrect** about 'A' and 'B'?

A) A and B are ionisation isomers.

B) 'B' can exhibit geometrical isomerism.

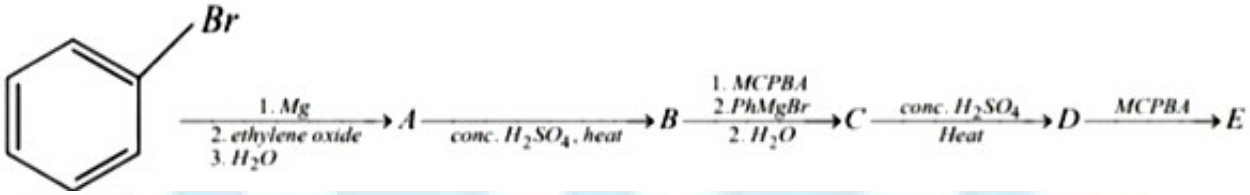
C) The spin only magnetic moment of 'A' is 3.87 B.M.

D) Both 'A' and 'B' can exhibit optical isomerism.

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.
The answer to each question is a **NUMERICAL VALUE.**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +2 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 43 and 44:



43. Molecular mass of final product E is _____

44. Degree of unsaturation in product D is _____

Question Stem for Question Nos. 45 and 46:

The standard EMF of the cell $Pt(s)\left|H_2(g)\right|HBr(aq)\left|AgBr(s)\right|Ag(s)$ was measured over a range of temperature and the data were fitted to the polynomial

$E^0=0.08-5\times10^{-4}\left(T-298\right)-3.45\times10^{-6}\left(T-298\right)^2$; [E^0 in Volts & T in Kelvin].

(Take $T=298K;\ F=96500C$)

45. Δ_rG^0 value (magnitude only, by considering n = 1) of the cell reaction $\left(in\ kJ\right)$ is

46. Δ_rS^0 value (magnitude only, by considering n = 1) of the cell reaction $\left(in\ JK^{-1}\right)$ is

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Question Stem for Question Nos. 47 and 48:

When a non-volatile solute (A) dissolves in water, it begins to dissociate slowly to establish following equilibrium.

$$A_{(aq)} \rightleftharpoons B_{(aq)} + C_{(aq)}$$

The progress of the reaction was followed determining the depression in freezing point. When 2 gm of A was dissolved in 500 gm of water the depression in freezing point was 0.2^0 at zero time and 0.25^0 after 30 minutes from the time of dissolution at which equilibrium established.

(Assume molality is equal to molarity, K_f of $H_2O = 2.0 K\,kg\,mol^{-1}$, $R = 0.08\,Lit.atm.mol^{-1}\,K^{-1}$) (Ignore variation of K w.r.t. T)

47. If equilibrium constant is $Y \times 10^{-3}$, The value of 'Y' is ------(up to 2 decimal places)

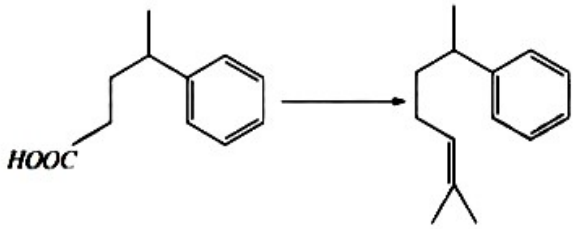
48. The Osmotic pressure of the above solution at 300K, at equilibrium (in atm.) is ----

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

49. Compound on the reactant side can be converted to compound on the product side (as the chief product) by which of the following sequence of reagents?



A) $i)\ LiAlH_4; ii) SOCl_2; iii)\ Mg / Et_2O; iv) Me_2CO; v) aq.NH_4Cl; vi) PCl_5; vii)\ EtOK / \Delta$

B) $i)\ SOCl_2; ii) (Me_2CH)_2 Cd;\ iii)\ NaBH_4 / EtOH;\ iv)\ SOCl_2;\ v)\ t-BuOK / \Delta$

C) $i)\ MeOH / H^+; ii)\ DlBAL-H / -78^0 C;\ iii)\ Me_2CHMgBr;\ iv)\ H_3O^+;$
$v) PCl_3; vi) KOH / EtOH$

D) $i)\ Me_2CHLi; ii)\ H_3O^+;\ iii)\ Al_2O_3 / \Delta$

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50. In the following reactions, K, L, M, and N are the major products.

$$PhCH(CH_3)CH_2CN \xrightarrow[(ii)\ CH_3MgCl,\ then\ aq.NH_4Cl]{(i)\ CH_3MgCl,\ then\ H_3O^+} K$$

$$PhCH_3 + PhCOCl \xrightarrow[(ii)\ CH_3MgCl,\ then\ aq.NH_4Cl]{(i)\ anhyd.\ AlCl_3} L$$

$$PhCH_2COCl \xrightarrow[(ii)\ CH_3MgCl,\ then\ aq.NH_4Cl]{(i)\ \frac{1}{2}\ Ph_2Cd} M$$

$$CH_3CH_2CHO \xrightarrow[(iii)\ HCN\ (iv)\ conc.H_2SO_4,\ \Delta]{(i)\ CH_3MgCl,\ then\ aq.\ NH_4Cl\ (ii)\ CrO_3,\ dil.H_2SO_4} N$$

The correct statement(s) about K, L, M and N is(are)

A) L has asymmetric carbon(s) and N has no asymmetric carbon(s)

B) Both K and N have asymmetric carbon(s)

C) L and M are not isomers

D) Both L and M have asymmetric carbon(s)

51. Which of the following are **incorrect** statements?

A) Tyndall effect can be observed from a colloidal solution with dispersed phase having the same refractive index as that of the dispersion medium.

B) Adsorption is accompanied by decrease in enthalpy and decrease in entropy of the system.

C) Coagulation value of the electrolytes $AlCl_3$ and NaCl for As_2S_3 sol are 0.093 and 52 respectively, then NaCl has 560 times greater coagulating power than $AlCl_3$.

D) Micelles are produced by dissolving a surfactant in any solvent at any temperature.

52. At 25^0C, the dissociation constant of a weak monoprotic acid, $HA(K_a)$ is numerically equal to the dissociation constant of its conjugate base, $A^-(K_b)$. Which of the following statement(s) is are **correct**?

A) The dissociation constant of the acid, HA, is 10^{-7} at 25^0C.

B) The pH of 0.1 M aqueous solution of the acid, HA, is 4.0

C) The pH of 0.1 M aqueous solution of the conjugate base (A^-) is 10.0

D) The pH of an aqueous solution containing 0.1 M HA and 0.01 M HC1 is nearly 2.0

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53. Which of the following are true?

A) From low grade copper ores, copper is extracted by hydrometallurgy (acid or bacteria leaching followed by reduction with scrap iron or H_2).

B) In the extraction of Aluminium, purified Al_2O_3 is mixed with Na_3AlF_6 or CaF_2 which lowers the melting point of the mixture and also increase conductivity.

C) Gallium, indium, germanium and silicon are all purified by zone refining

D) In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used respectively are O_2 and Zn dust.

54. $\text{Test salt} \xrightarrow[\text{warm}]{\text{dilute } H_2SO_4} X\uparrow + \text{other products}$

Then identify the correct statements?

A) If 'X is a colourless, odourless gas with brisk effervescence, which turns lime water milky then possible anion present in the test salt is CO_3^{2-} (carbonate)

B) If 'X' is colourless gas with the smell of rotten eggs which turns lead acetate paper black, then possible anion present in the test salt is S^{2-} (sulphide)

C) If 'X' is a colourless gas with a purgent smell like burning sulphur which turns acidified dichromate solution green, then possible anion present in the test salt is SO_3^{2-} (sulphite)

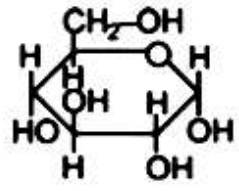
D) If 'X' is brown in colour which turn acidified KI solution containing starch solution blue, then possible anion present in the test salt in NO_3^- (nitrate)

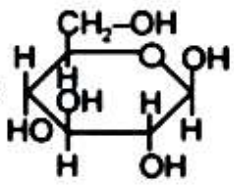
SECTION - 4 (Maximum Marks : 12)

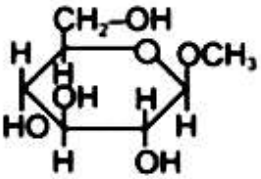
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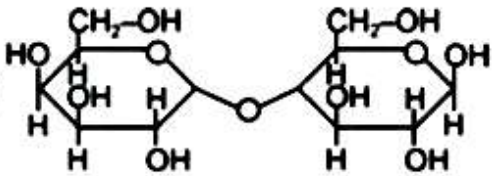
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

55. From given compounds

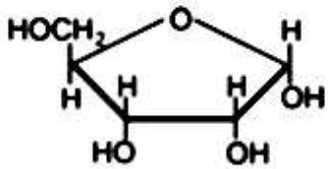
(A) 

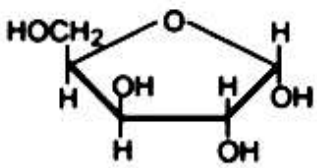
(B) 

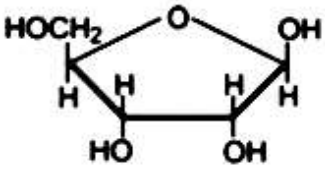
(C) 

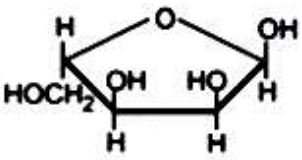
(D) 

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(E) 

(F) 

(G) 

(H) 

From given structures

Total number of reducing sugar $= x$

Total number of sugars in which molecular weight increases by 210, whenever it undergo acetylation reaction with excess of acetic anhydride $= y$

Then find the value of $x - y = ?$

56.

$$CH_3MgBr + \overset{*}{C}O_2 \rightarrow A \xrightarrow{H^+} B \xrightarrow{SOCl_2} C \xrightarrow{NH_3} E \xrightarrow{Br^-O^-} G$$

$$C \xrightarrow[Pd-BaSO_4]{H_2} D \xrightarrow{NaOI} F\,(ppt) + H$$

The number of correct statements related to this sequential reaction is

I) B is an acid can release CO_2 from $NaHCO_3$

II) Both E and F have isotopic carbon as specified in initial $*CO_2$

III) F by reaction with silver powder release a gaseous product which form white precipitate with tollen's reagent

IV) Calcium salt of 'H' by dry distillation gives acetone as one of the products

V) G gives yellow colour oily liquid with HNO_2

VI) C by reaction with $(CH_3)_2Cd$ form a compound which gives yellow precipitate (ppt) with $NaOI$

57. The total number of isomers for a square planar complex $[M(F)(Cl)(SCN)(NO_2)]$ is

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